Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/29/2006
PI Name:	Mount, Frances E Ph.D.		
Project Title:	Human-Systems Design Requirements Data Base and Tracking System		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Space human factors engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	NASA CENTER	**Phone:**	281-244-0117
Organization Name:	Johnson Space Center		
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PI Address 2:	2101 NASA Road One		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77058-3696	**Congressional District:**	22
Comments:	What is COI info??		
Project Type:	Ground	**Solicitation / Funding Source:**	INTERNAL PROJECT
Start Date:	12/04/2000	**End Date:**	03/31/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Booher,Cletis		
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The Space Human Factors Engineering (SHFE) Program at the National Aeronautics and Space Administration's (NASA's) Johnson Space Center (JSC) in Houston, TX, has identified a number of "Critical Questions" related to human factors as applied to longer duration missions. These questions need answers before long-duration missions can be planned. For example, Critical Question 3.1 asks, "What human systems design requirements are necessary to maintain human-system performance during all phases of the mission, including all anticipated changes in crew performance?" Future human-based space missions may have durations many times longer than the currently anticipated International Space Station (ISS) missions, and communications with the ground may be very limited. For long-duration missions, the level of isolation and autonomy will be far greater than in any previous space missions. Equipment and habitat design, supplies, training materials, and crew operations must be planned on the basis of the best available information from numerous disciplines, including human factors, biomechanics, education, cognitive and		

	social psychology, and physiology. New technologies developing before the mission must be tracked and their human interfaces understood. Automation and intelligent systems will play a major role in this type of mission, and the allocation of tasks, responsibilities, and time to various players – human or machine – must be based on the best information available. The SHFE Program will maintain a database of the critical questions, associated research and the authors of related papers.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	As with all space-related research, the information developed to support exploration-class missions will result in direct benefits to earth-bound programs and problems. Areas of endeavor which will benefit directly from this research include exploration of other extreme environments and long-duration processes and missions of many types.
Task Progress:	No progress report this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)